Block Cipher Principles

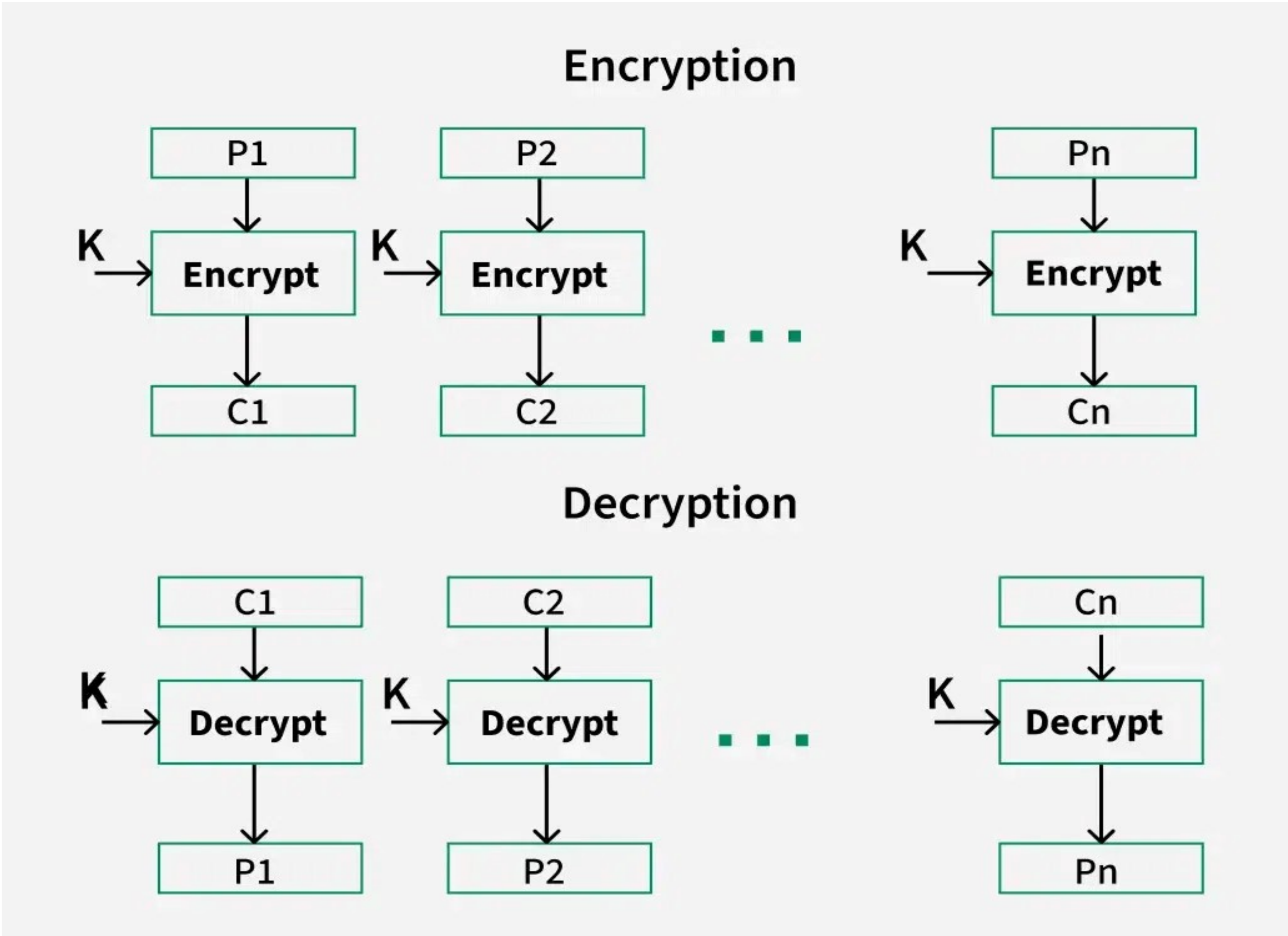
Block ciphers are built in the Feistel structure.

- Block cipher has a specific number of rounds and keys for generating ciphertext.Block cipher is a type of encryption algorithm that processes fixed-size blocks of data, usually 64 or 128 bits, to produce ciphertext.
- Block cipher operates on fixed length of Blocks and uses symmetric key.

Modes of Block Cipher

- Electronic Code book
- Cipher block chaining
- Cipher Feedbck Mode
- Output feedback mode
- Counter Mode

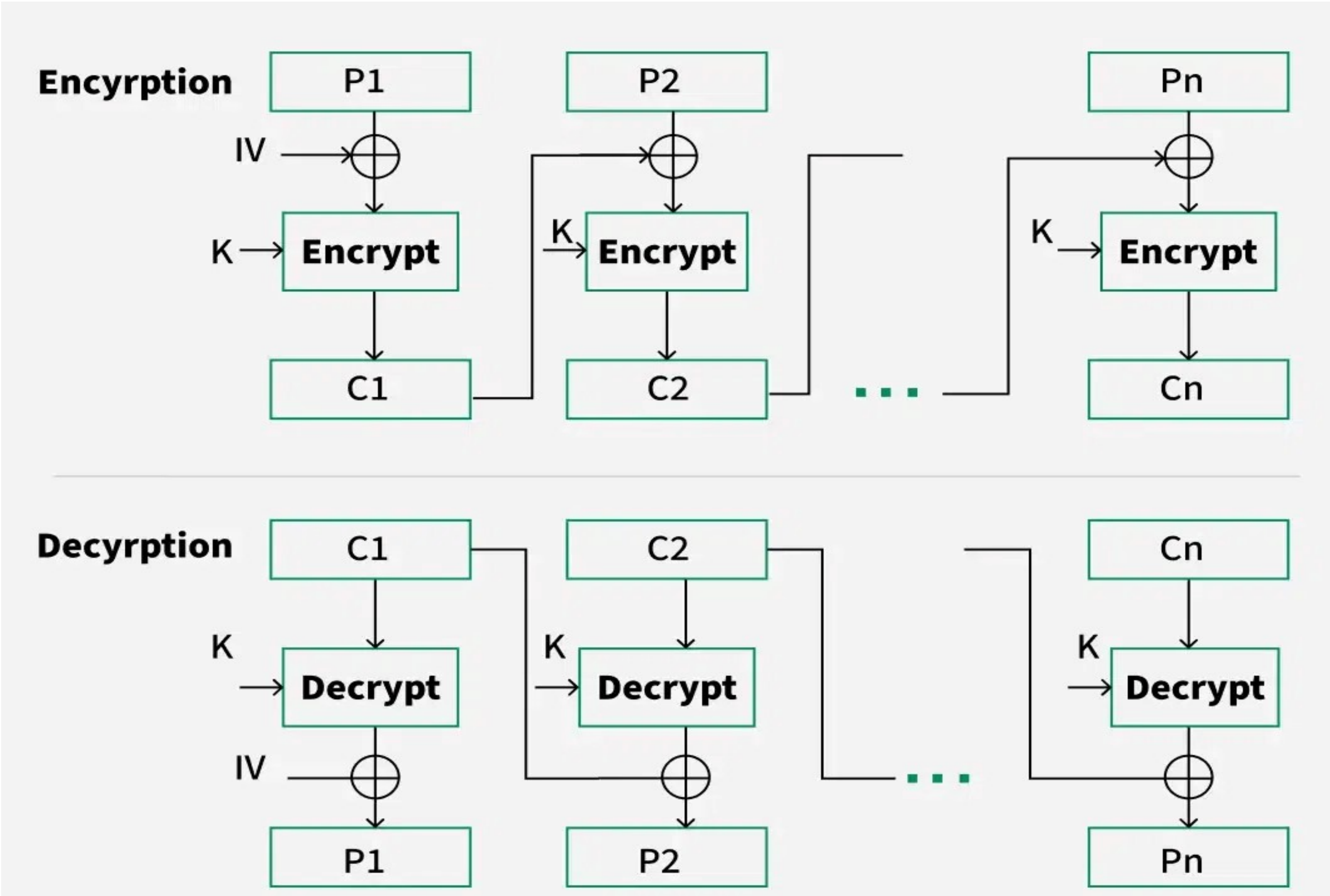
Electronic Code Book (ECB)



- **Advantages of using ECB**
  Parallel encryption of blocks of bits is possible, thus it is a faster way of encryption.

  Simple way of the block cipher.

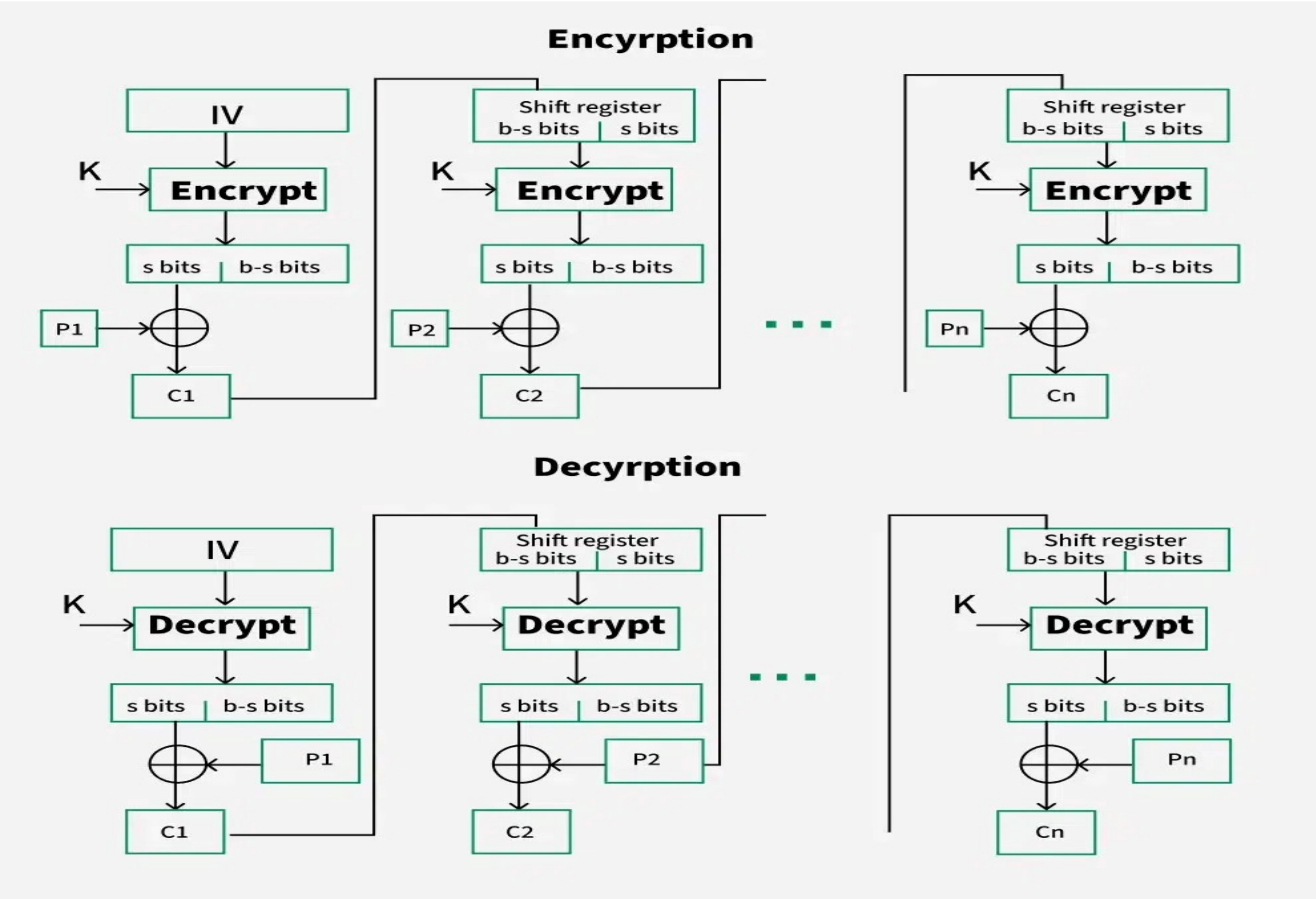
Cipher Block Chaining



Advantages of CBC

- CBC is a good authentication mechanism.
- Better resistive nature towards cryptanalysis than ECB.
- More secure than ECB as it hides patterns.

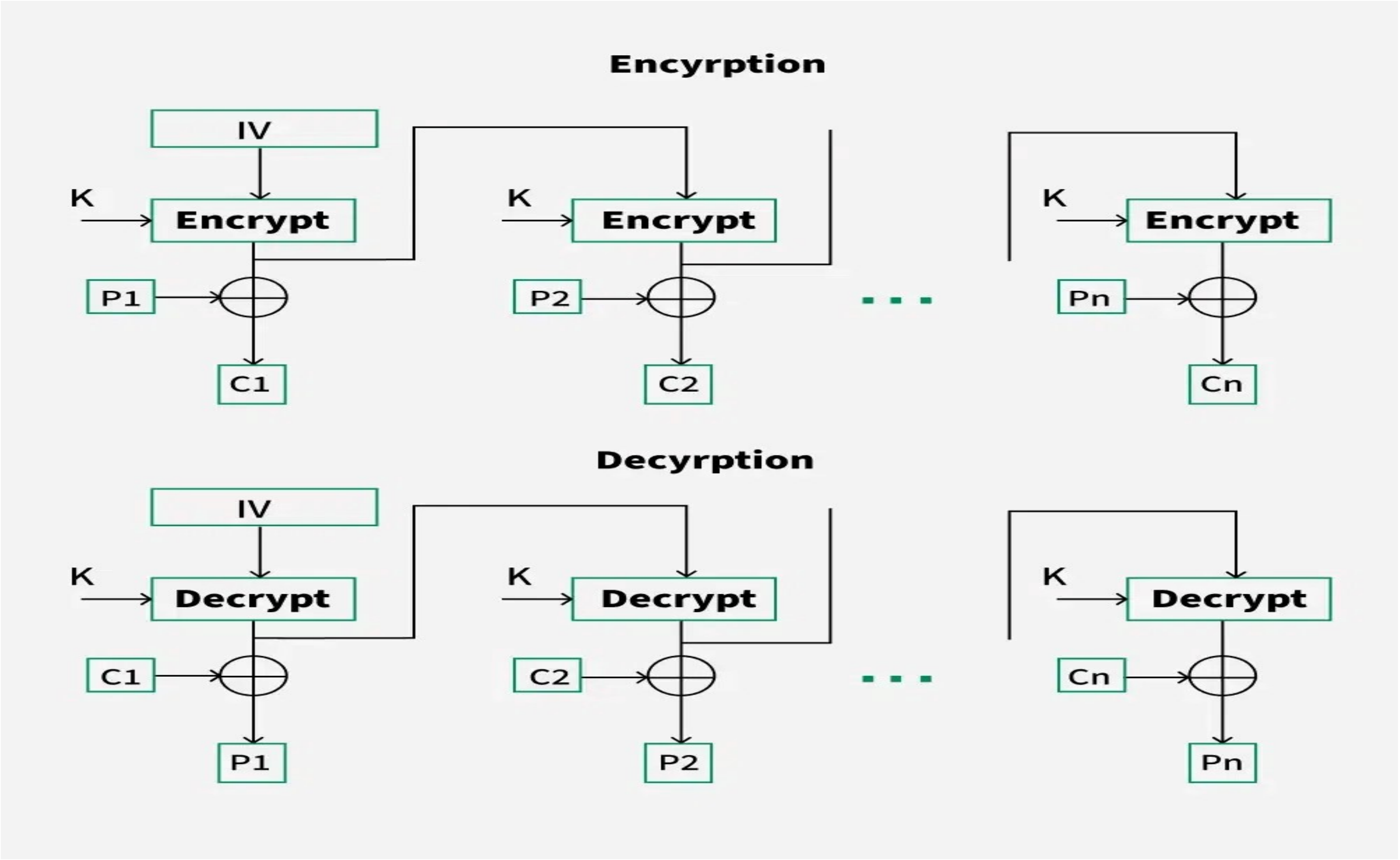
Cipher Feedback Mode (CFB)



Advantages of CFB

- Since, there is some data loss due to the use of shift register, thus it is difficult for applying cryptanalysis.
- Can handle data streams of any size

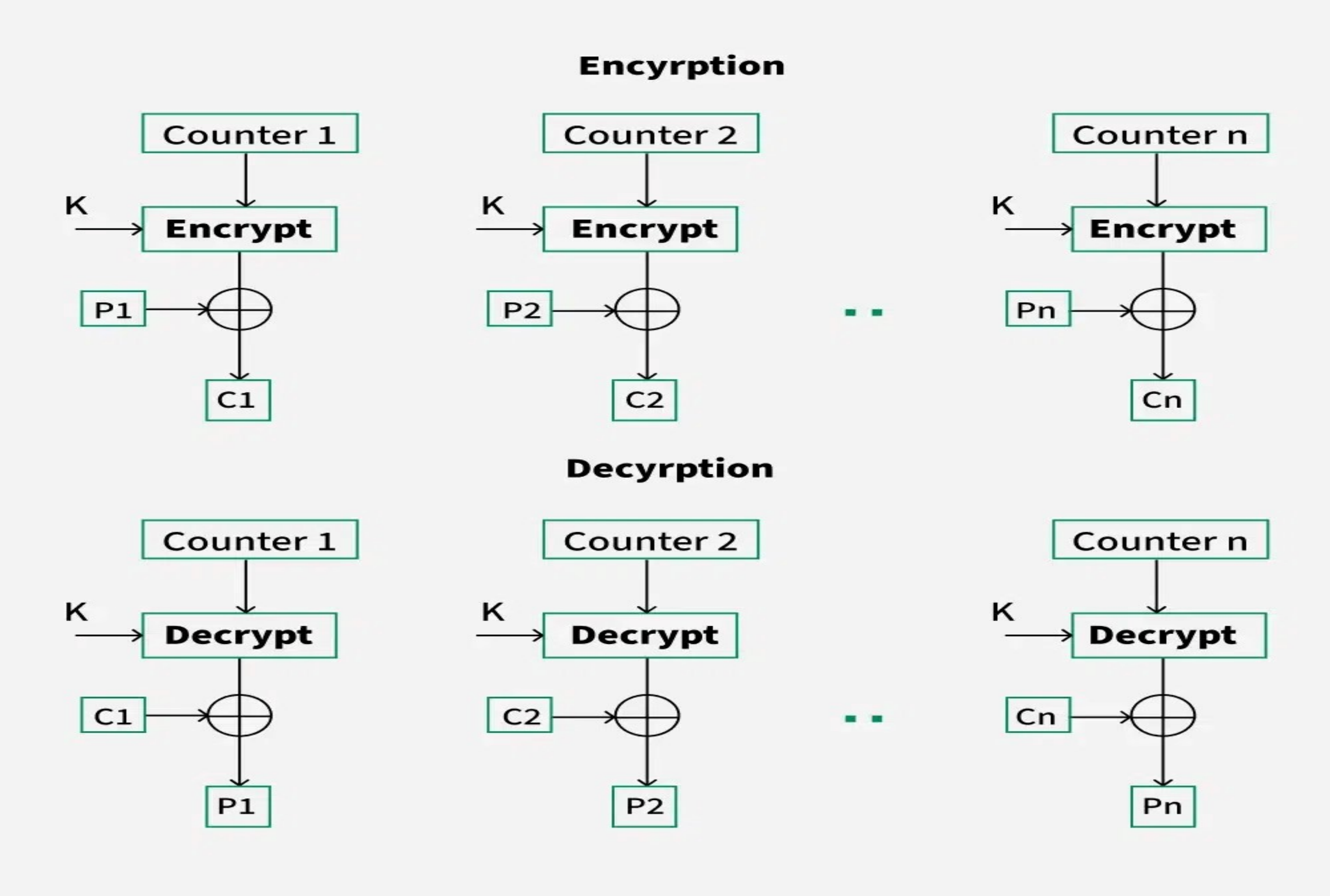
Output Feedback Mode



Advantages of OFB

In the case of CFB, a single bit error in a block is propagated to all subsequent blocks. This problem is solved by OFB as it is free from bit errors in the plaintext block. Thus errors in transmission don’t propagate.

Counter Mode



Advantages of Counter

Since there is a different counter value for each block, the direct plaintext and ciphertext relationship is avoided. This means that the same plain text can map to different ciphertext.

Parallel execution of encryption is possible as outputs from previous stages are not chained as in the case of CBC.